

PUERTO BARRIOS, Guatemala

WMO: 786370

Lat: 15.731N

Lon: 88.584W

Elev: 10

StdP: 101.20

Time Zone: -6.00 (W06)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	18.1	15.0	10.7	19.7	16.1	11.4	20.3	8.4	22.8	7.3	23.2	0.7	270	0.272

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	7.4	33.8	27.1	33.0	27.0	32.2	26.7	28.2	32.0	27.7	31.5	27.3	31.1	3.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	31.1	26.8	22.5	30.7	26.2	21.6	29.8	90.3	32.2	88.4	31.8	86.7	31.5	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.6	5.0	15.0	36.3	0.9	1.4	14.4	37.3	13.9	38.1	13.4	38.9	12.8	39.9		
(5)			14.4	29.2	1.1	0.5	13.6	29.6	13.0	29.9	12.3	30.2	11.5	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	23.4	24.4	25.4	27.0	27.8	27.8	27.5	27.6	27.7	26.7	24.8	23.9		
	DBStd	2.17	1.86	1.74	1.87	1.66	1.28	1.08	0.84	0.74	0.97	1.45	1.60	1.72		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	5903	416	403	477	511	551	534	542	547	532	518	443	429		
	CDD18.3	2861	157	170	219	261	292	284	284	288	282	259	193	171		
	CDH23.3	25218	796	1086	1719	2546	3087	2998	2857	2956	2853	2253	1167	901		
(13)	CDH26.7	8613	143	274	553	971	1206	1120	1016	1108	1067	723	259	173		
(14)	Wind	WSAvg	1.5	1.7	1.6	1.7	1.6	1.6	1.8	1.5	1.4	1.4	1.3	1.3		
(15) Precipitation	PrecAvg	3031	237	146	122	144	144	267	435	384	276	315	298	251		
	PrecMax	4170	495	546	391	318	346	461	896	684	550	514	768	694		
	PrecMin	2022	20	26	8	26	24	137	205	220	85	57	156	73		
	PrecStd	508	116	110	85	88	79	84	172	134	110	132	138	145		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	31.8	33.9	35.1	34.2	34.1	33.0	33.2	34.1	33.2	31.2	30.3		
		MCWB	24.9	25.5	25.7	26.7	27.2	27.4	26.9	27.1	26.9	26.8	25.9	25.9		
	2%	DB	29.0	30.1	31.9	33.1	33.1	33.0	32.1	32.2	33.0	32.0	30.1	29.2		
		MCWB	24.7	25.3	26.3	27.1	27.2	27.3	26.6	26.9	27.0	26.5	25.7	25.2		
	5%	DB	28.1	29.2	30.8	32.0	32.1	32.0	31.2	31.9	32.1	31.1	29.1	28.3		
		MCWB	24.3	24.8	26.1	26.6	26.9	26.8	26.6	26.8	26.7	26.5	25.0	24.5		
	10%	DB	27.2	28.4	29.8	31.0	31.2	31.1	30.9	31.1	31.2	30.2	28.1	27.8		
		MCWB	23.6	24.4	25.2	26.3	26.6	26.7	26.6	26.7	26.7	25.9	24.2	24.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.9	26.7	27.8	28.6	28.7	28.6	28.0	28.2	28.1	27.9	26.9	26.8		
		MCDB	28.9	30.2	31.5	33.1	32.6	32.5	31.4	31.8	32.2	31.6	30.1	29.7		
	2%	WB	25.2	25.9	26.8	27.7	28.0	27.9	27.2	27.6	27.5	27.2	26.2	25.8		
		MCDB	28.2	29.4	30.8	31.7	31.8	31.6	30.8	31.3	31.5	30.8	29.2	28.7		
	5%	WB	24.7	25.3	26.2	27.1	27.4	27.3	26.9	27.1	27.1	26.7	25.6	25.1		
		MCDB	27.6	28.6	29.8	30.9	31.1	30.9	30.5	30.8	31.0	30.1	28.4	27.9		
	10%	WB	24.1	24.7	25.5	26.6	26.9	26.9	26.6	26.7	26.7	26.3	25.0	24.5		
		MCDB	26.7	27.8	28.8	30.1	30.5	30.4	29.9	30.2	30.3	29.4	27.6	27.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.5	8.3	8.4	7.8	7.4	7.2	7.6	7.6	6.9	6.8	6.7		
		MCDBR	8.3	8.7	9.5	9.3	8.5	8.4	7.7	8.3	8.6	8.1	7.8	7.9		
		MCWBR	4.6	4.6	4.7	4.3	3.9	3.8	3.6	4.0	3.7	3.8	3.9	4.4		
	5% WB	MCDBR	7.8	8.3	8.9	8.6	7.9	7.7	7.4	7.9	8.0	7.5	7.1	7.4		
		MCWBR	4.6	4.7	4.7	4.3	3.9	3.9	3.7	3.9	3.8	3.8	3.9	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.382	0.395	0.429	0.510	0.559	0.474	0.487	0.459	0.437	0.426	0.401	0.403		
	taud		2.523	2.482	2.369	2.126	2.002	2.291	2.234	2.326	2.378	2.431	2.487	2.469		
	Ebn at Noon		910	913	887	813	762	821	812	843	867	871	883	877		
	Edh at Noon		102	110	126	161	180	133	142	130	123	114	104	105		
(44) All-Sky Solar Radiation	RadAvg		3.66	4.65	5.38	5.76	5.36	5.26	5.38	5.47	5.27	4.19	3.63	3.56		
	RadStd		0.42	0.40	0.41	0.29	0.35	0.38	0.23	0.28	0.23	0.46	0.25	0.34		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page